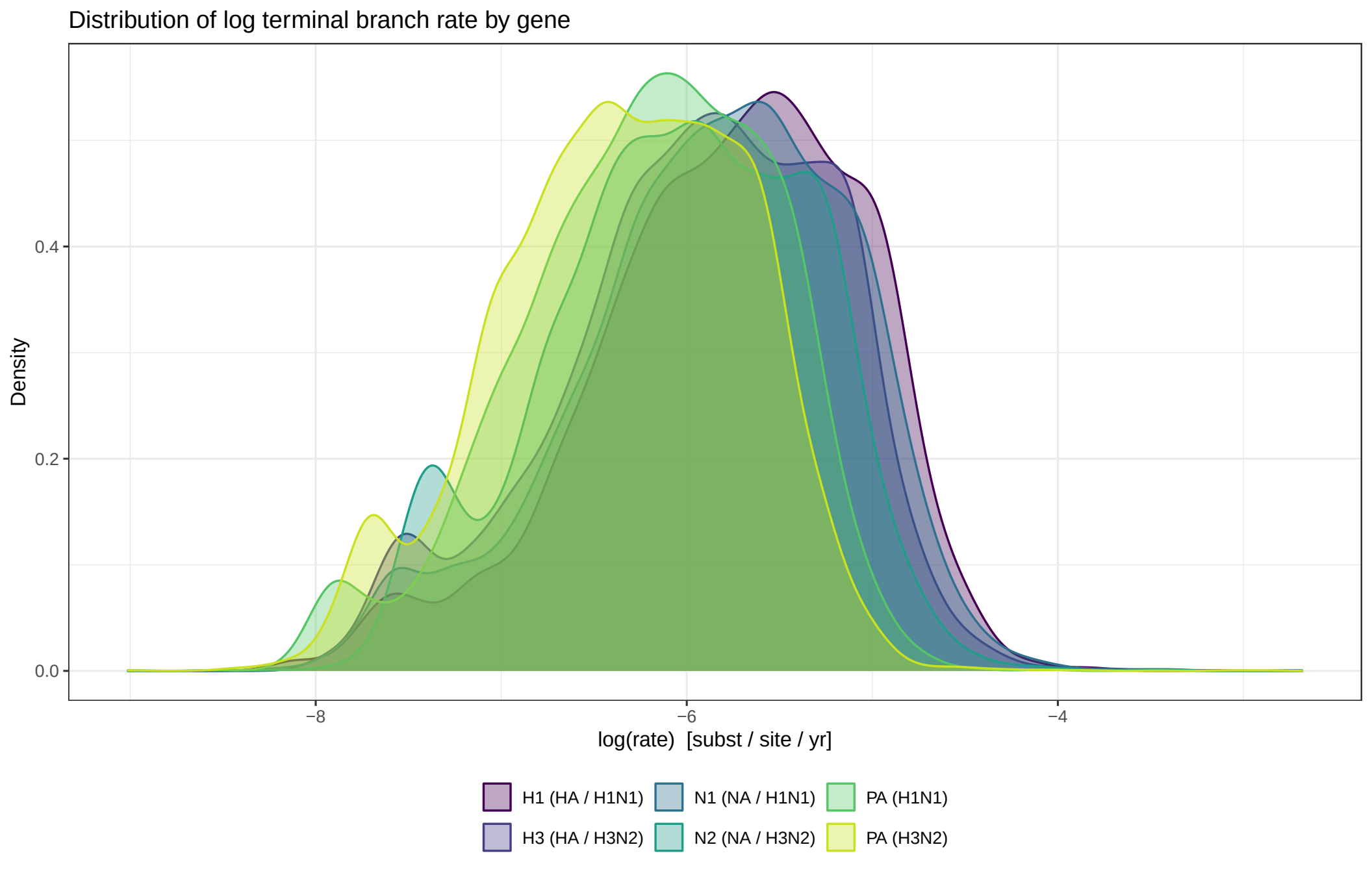
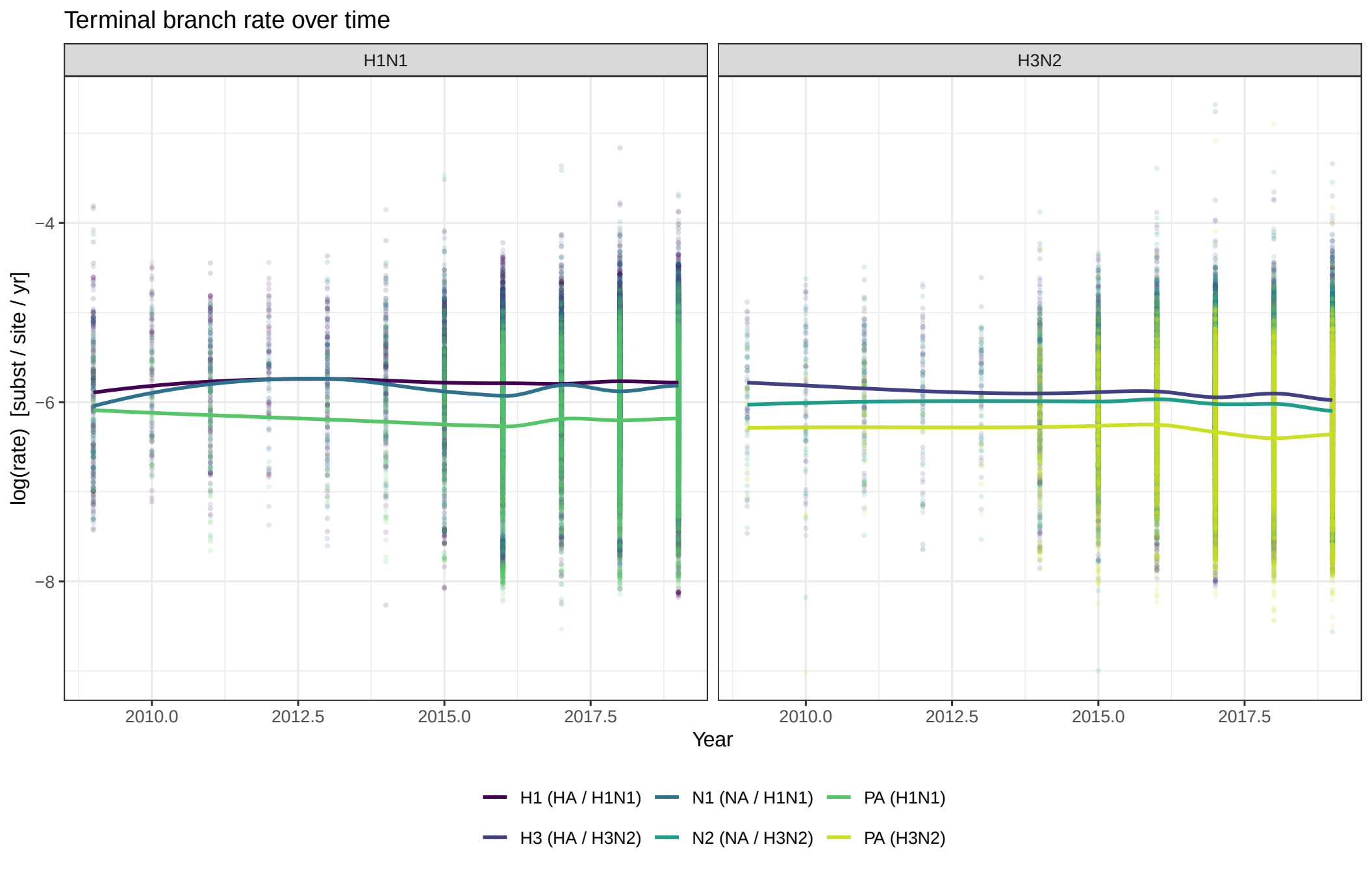
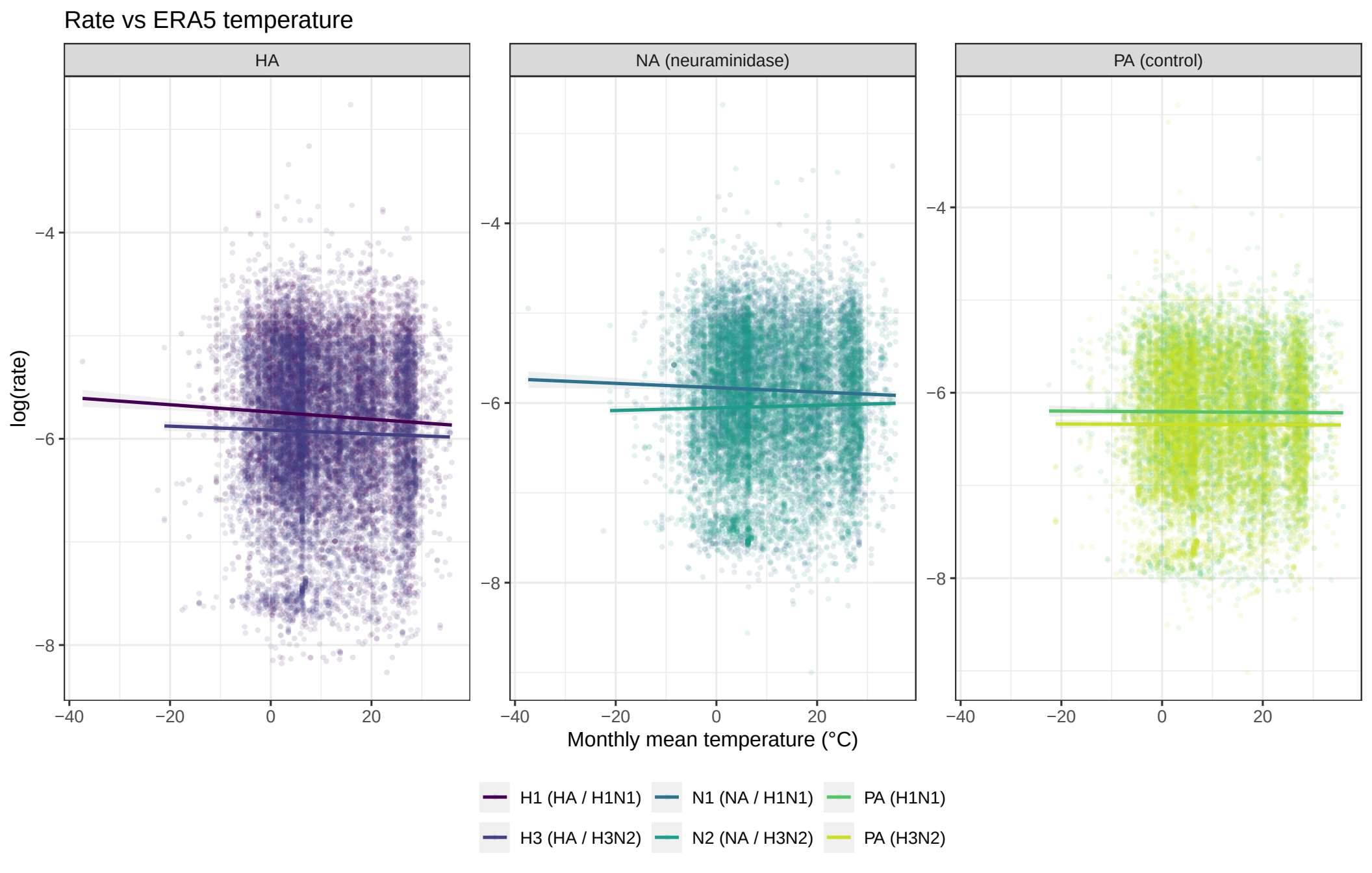
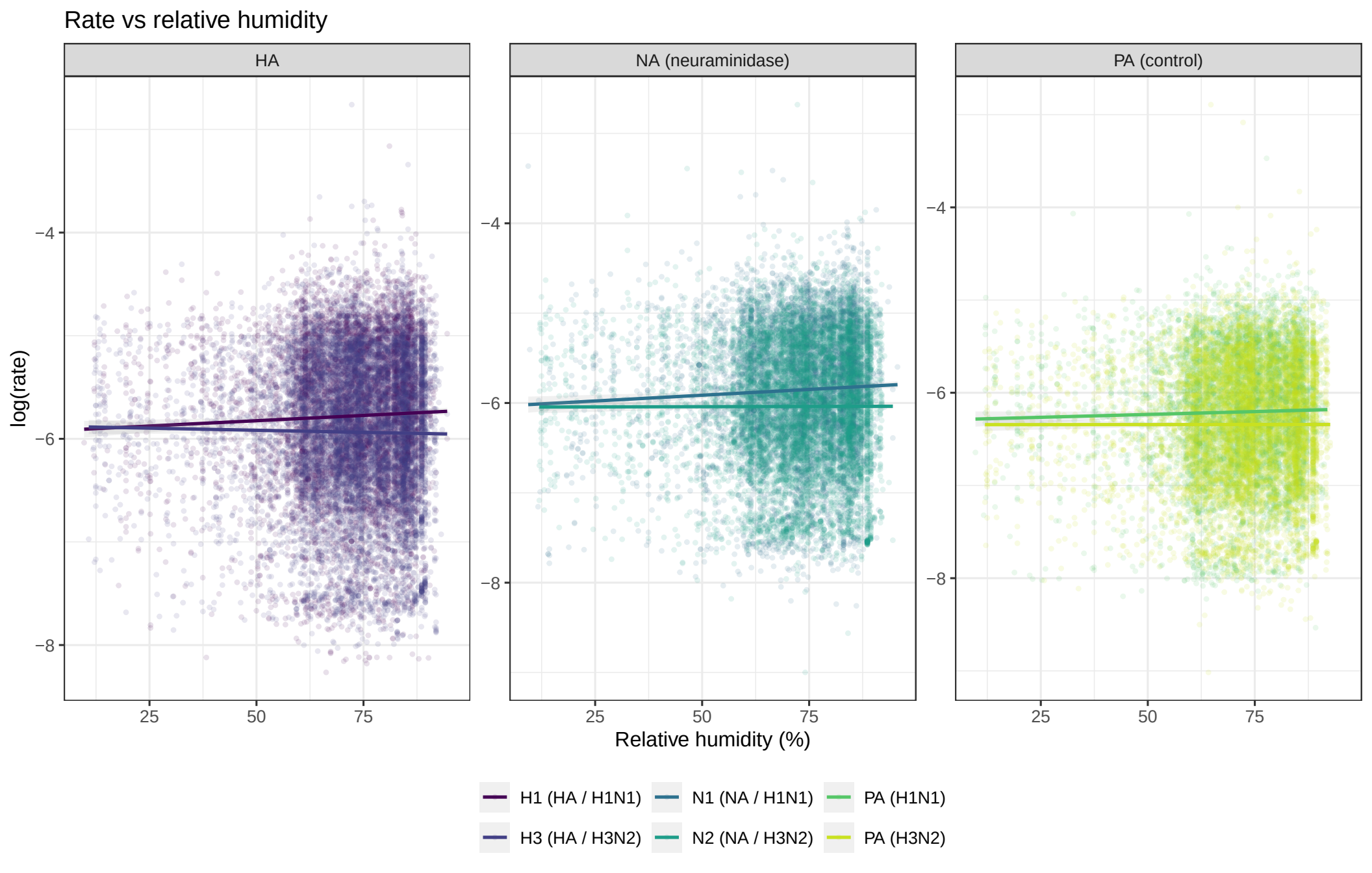
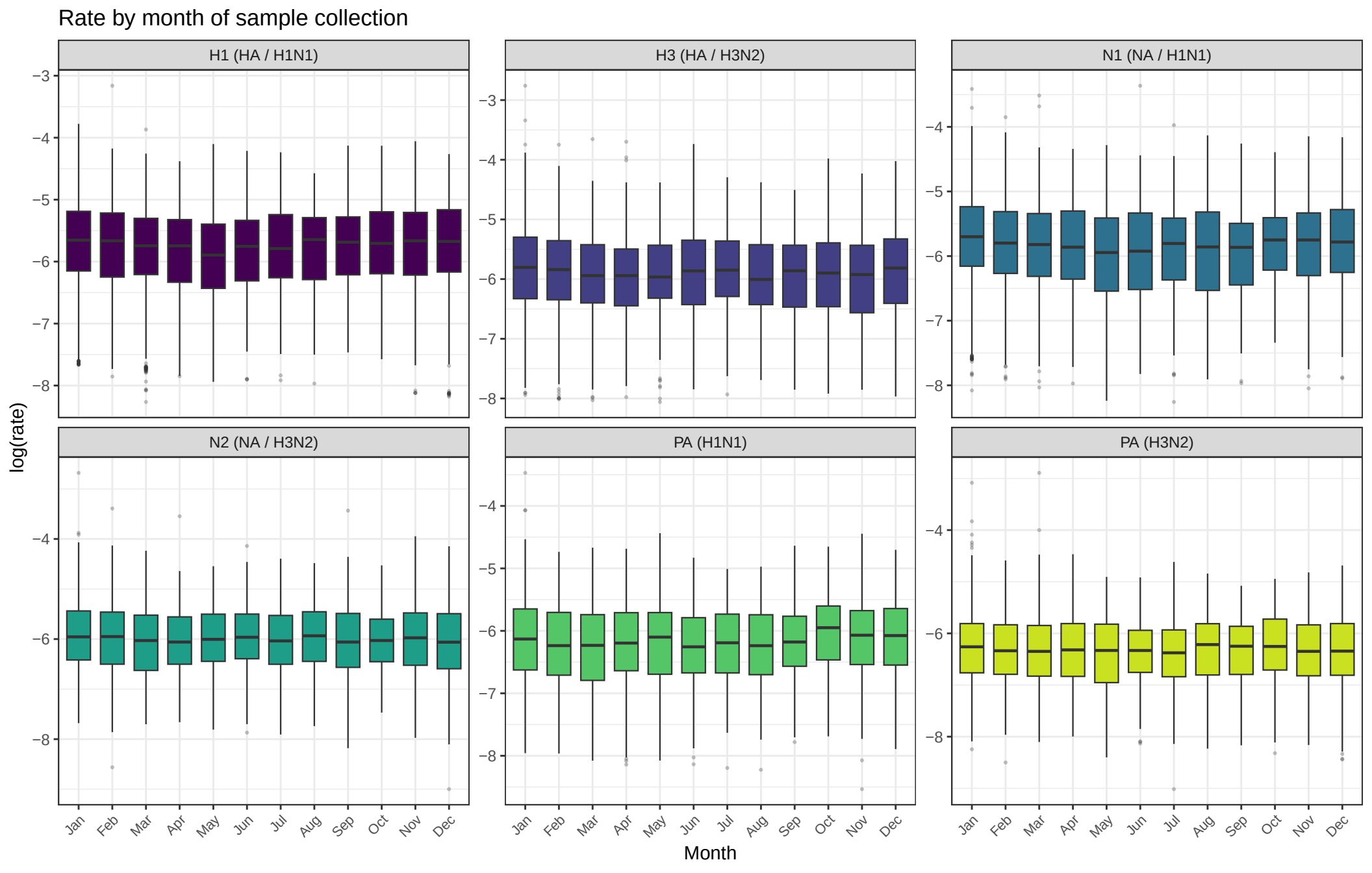
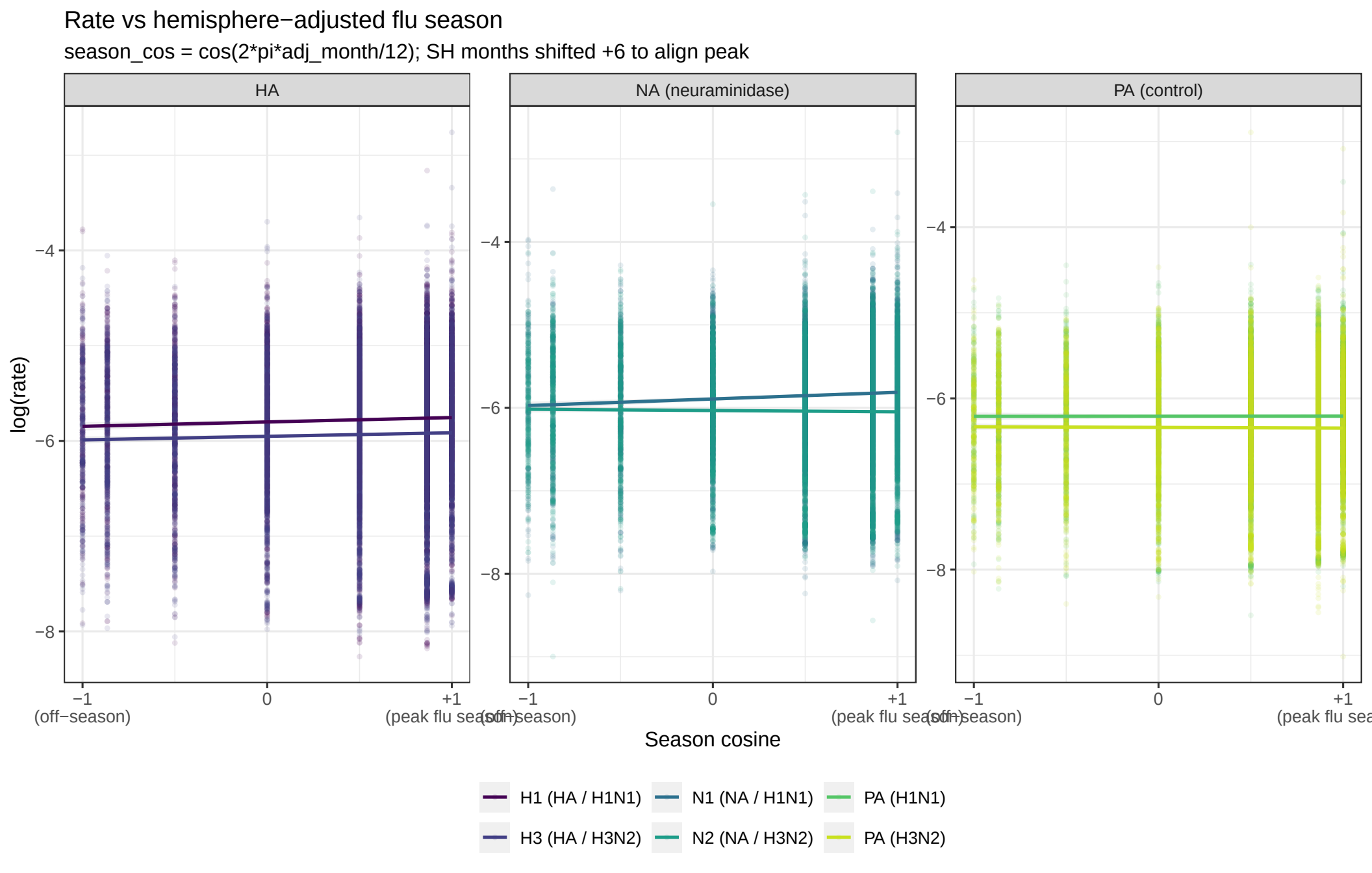

Distribution of log terminal branch rate by gene
Density
0.0
0.2
0.4
−8
−6
−4
log(rate) [subst / site / yr]
H1 (HA / H1N1)
N1 (NA / H1N1)
PA (H1N1)
H3 (HA / H3N2)
N2 (NA / H3N2)
PA (H3N2)

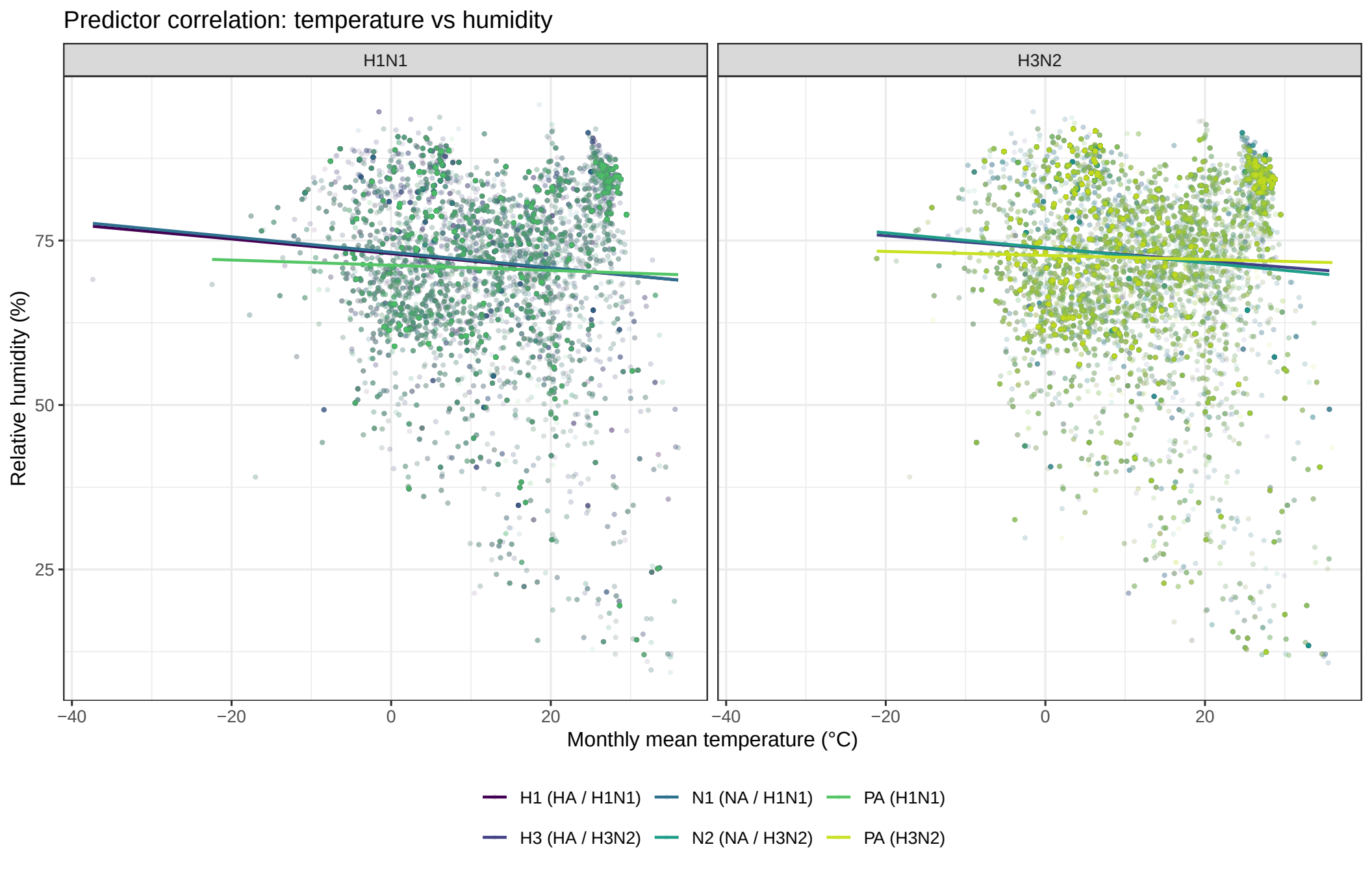

Terminal branch rate over time
H1N1
H3N2
log(rate) [subst / site / yr]
−4
−6
−8
2010.0
2012.5
2015.0
2017.5
Year
H1 (HA / H1N1)
N1 (NA / H1N1)
PA (H1N1)
H3 (HA / H3N2)
N2 (NA / H3N2)
PA (H3N2)


Rate vs ERA5 temperature
HA
NA (neuraminidase)
PA (control)
log(rate)
−4
−6
−8
−40
−20
0
20
Monthly mean temperature (°C)
H1 (HA / H1N1)
N1 (NA / H1N1)
PA (H1N1)
H3 (HA / H3N2)
N2 (NA / H3N2)
PA (H3N2)

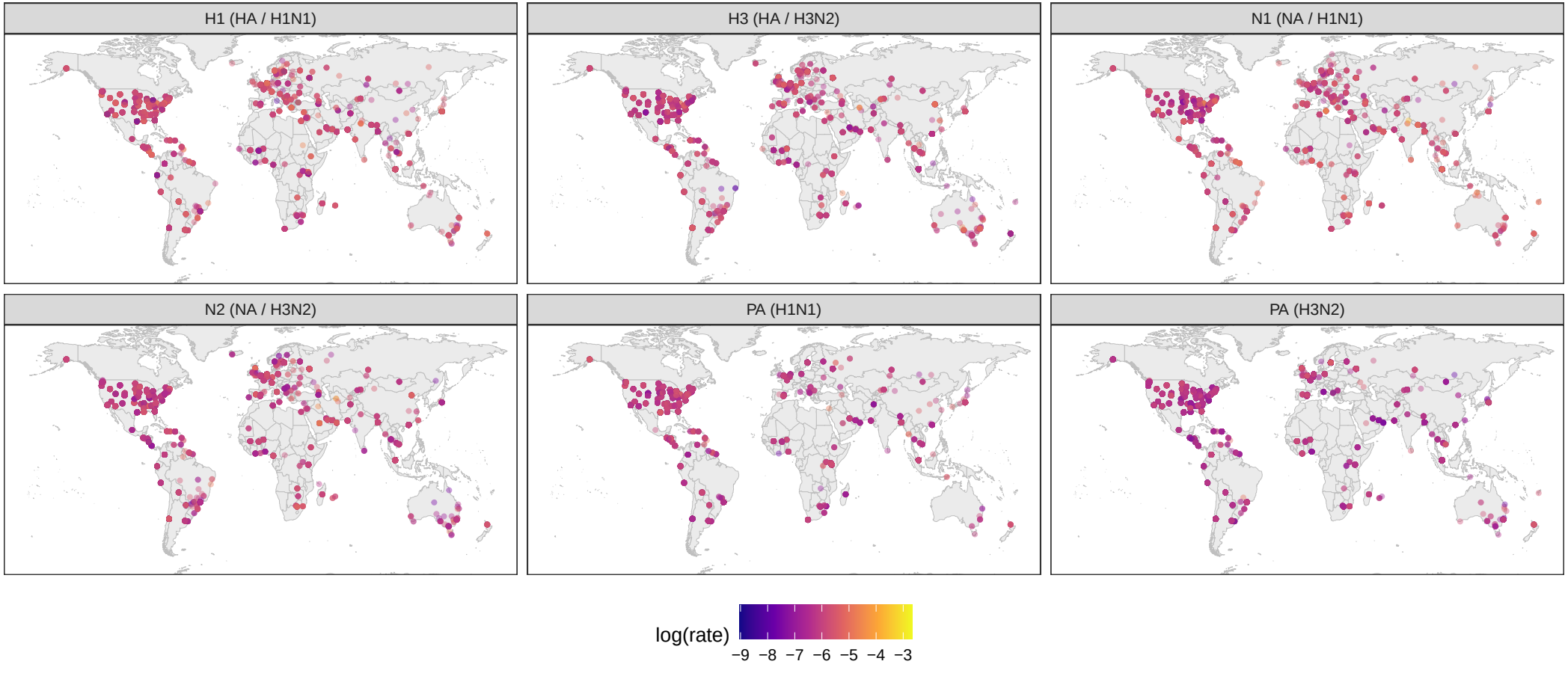

Rate vs relative humidity
HA
NA (neuraminidase)
PA (control)
log(rate)
−4
−6
−8
25
50
75
Relative humidity (%)
H1 (HA / H1N1)
N1 (NA / H1N1)
PA (H1N1)
H3 (HA / H3N2)
N2 (NA / H3N2)
PA (H3N2)

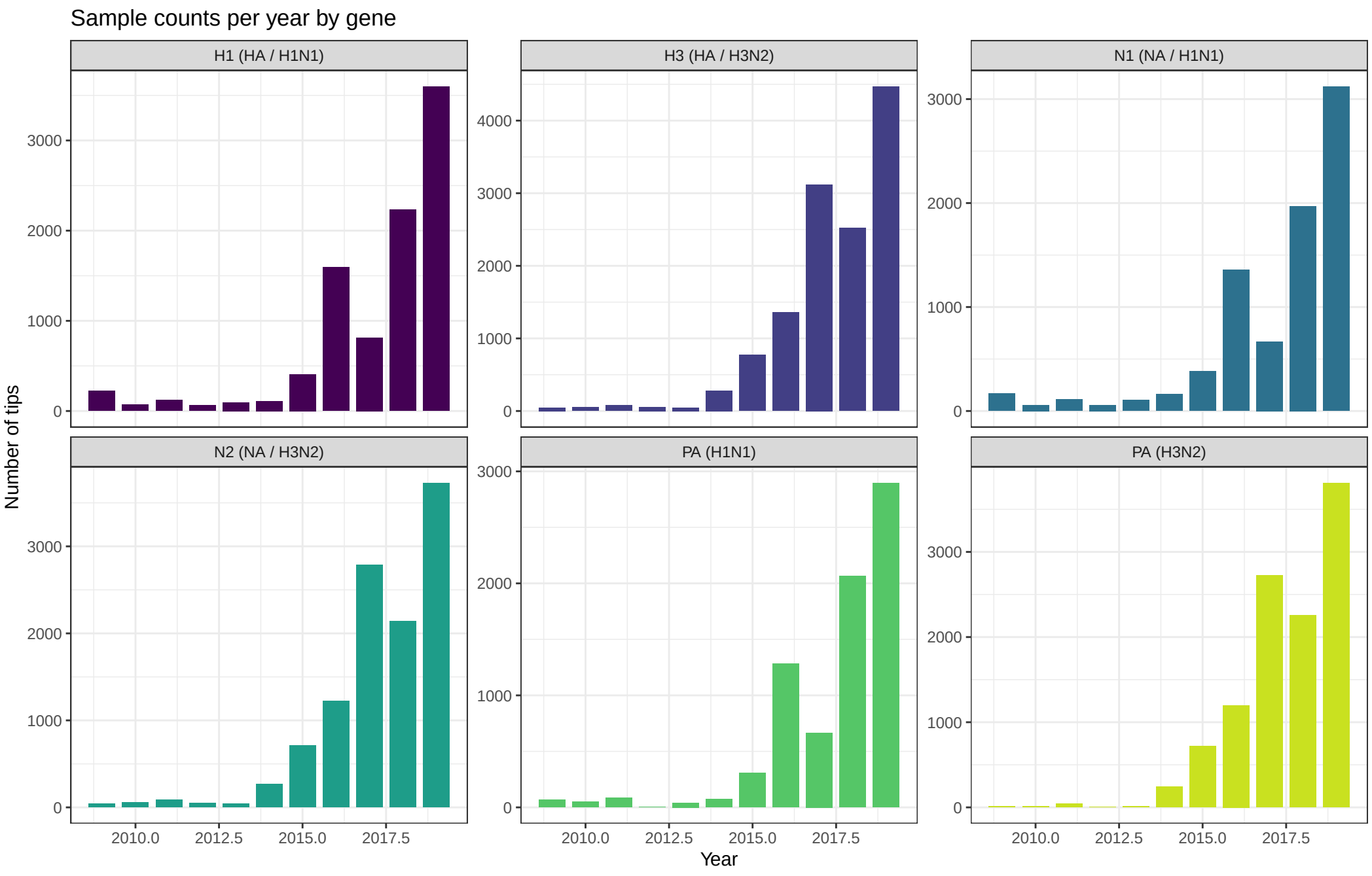

Rate by month of sample collection
H1 (HA / H1N1)
H3 (HA / H3N2)
N1 (NA / H1N1)
N2 (NA / H3N2)
PA (H1N1)
PA (H3N2)
log(rate)
−3
−4
−5
−6
−7
−8
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Month

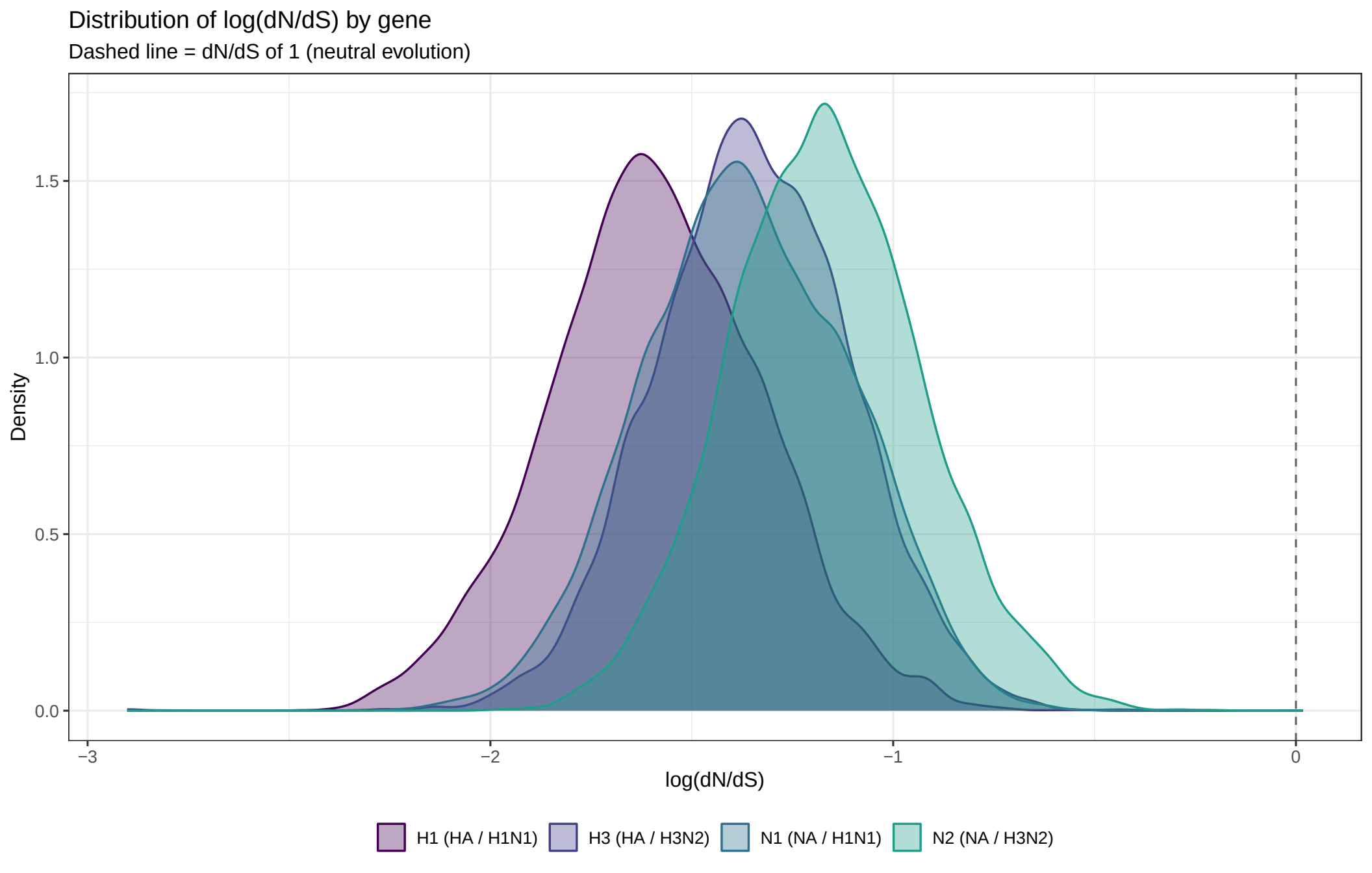

Rate vs hemisphere−adjusted flu season
season_cos = cos(2*pi*adj_month/12); SH months shifted +6 to align peak
HA
NA (neuraminidase)
PA (control)
log(rate)
−4
−6
−8
−1
0
+1
(off−season)
(peak flu season)
Season cosine
H1 (HA / H1N1)
N1 (NA / H1N1)
PA (H1N1)
H3 (HA / H3N2)
N2 (NA / H3N2)
PA (H3N2)

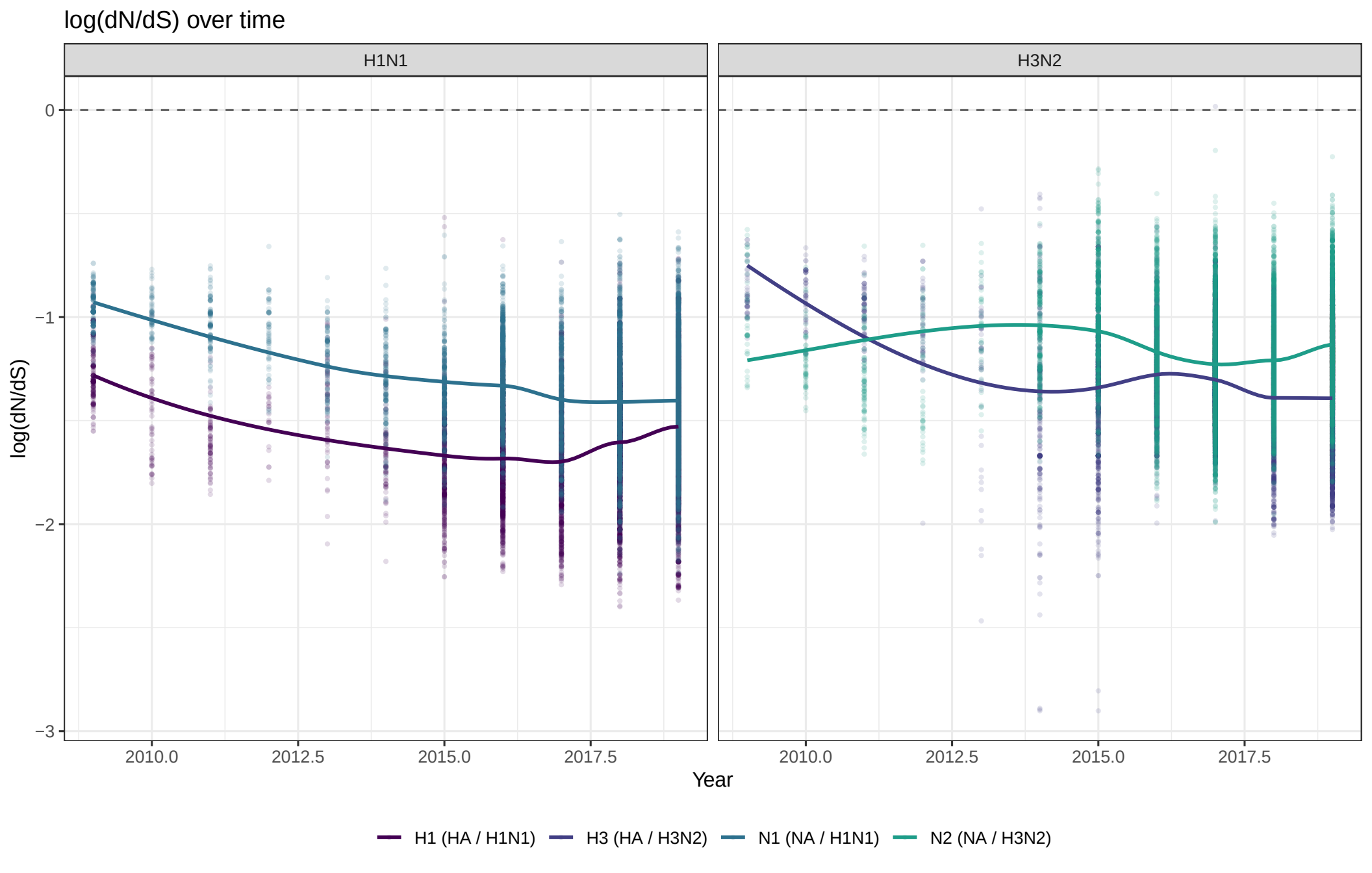

Predictor correlation: temperature vs humidity
H1N1
H3N2
Relative humidity (%)
75
50
25
−40
−20
0
20
Monthly mean temperature (°C)
H1 (HA / H1N1)
N1 (NA / H1N1)
PA (H1N1)
H3 (HA / H3N2)
N2 (NA / H3N2)
PA (H3N2)

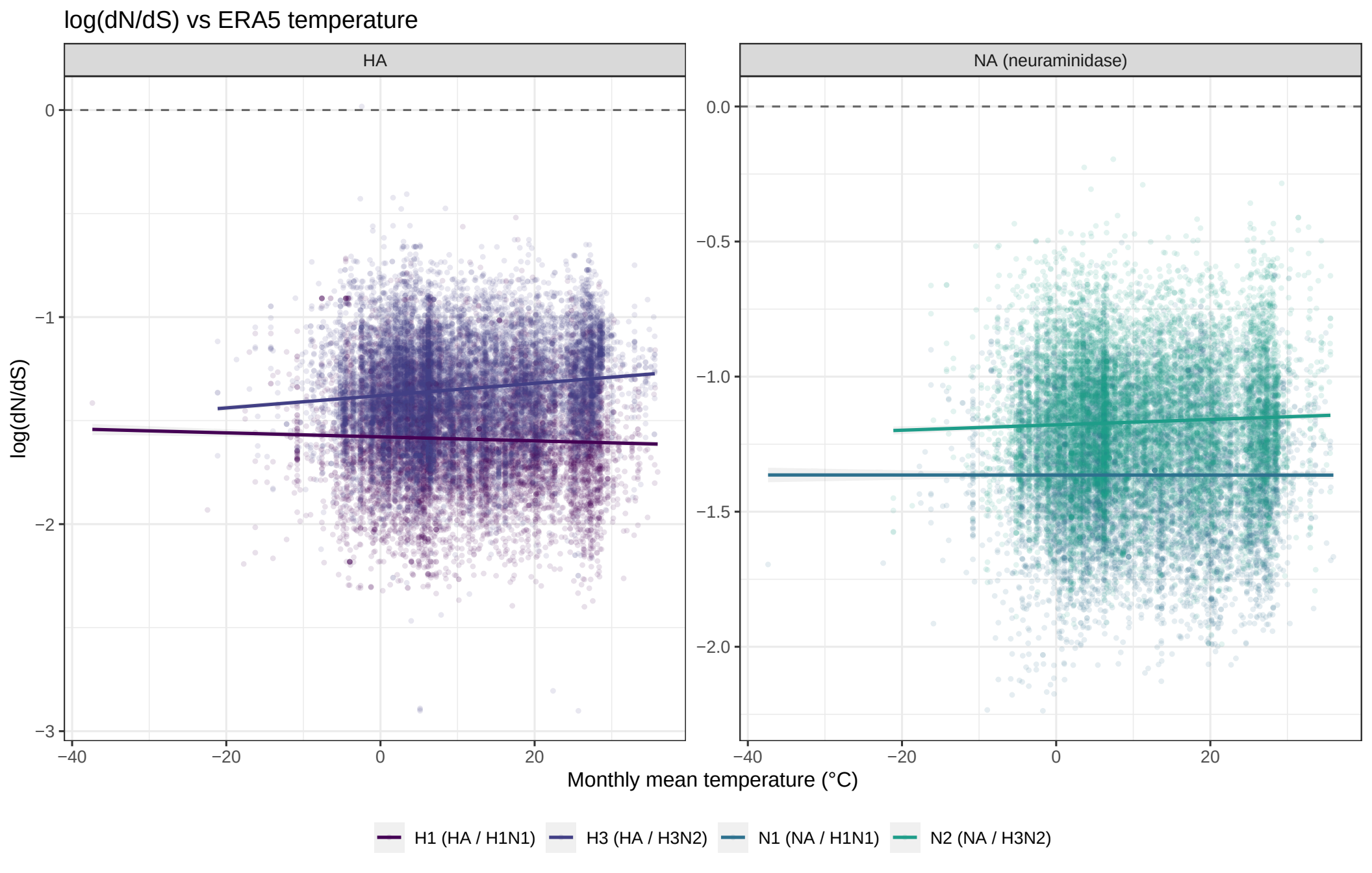
Geographic distribution of terminal branch rates

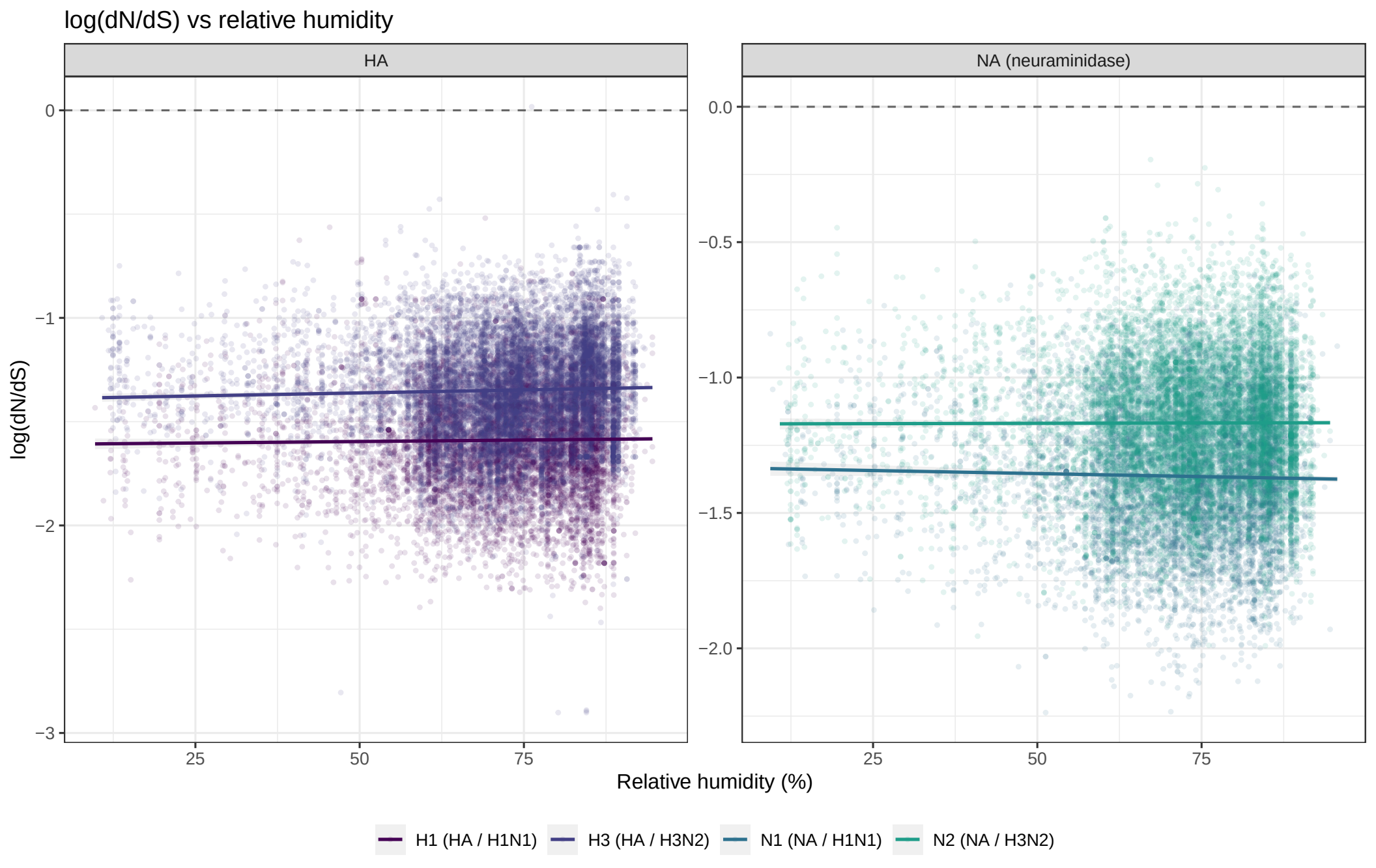


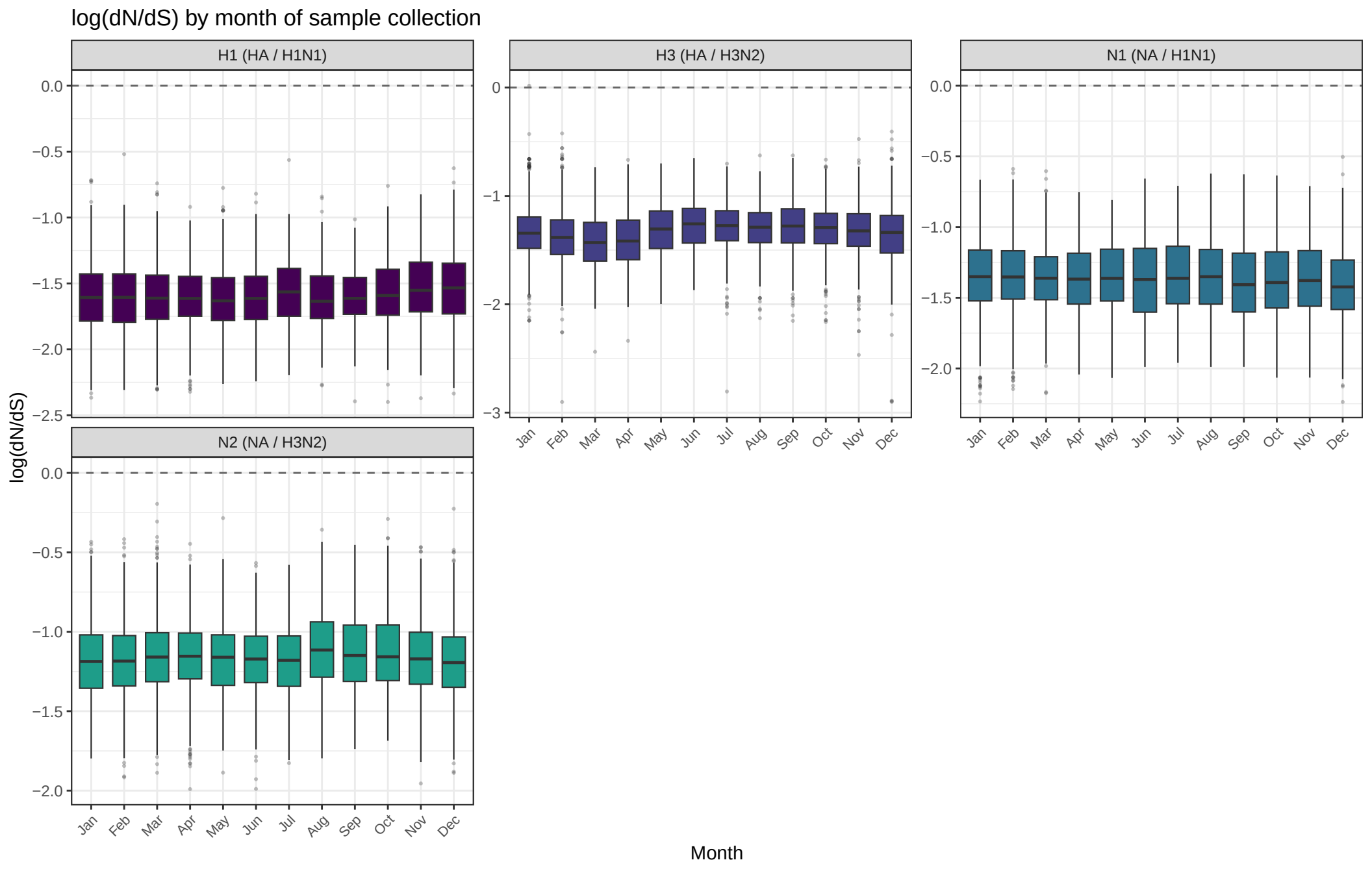

Sample counts per year by gene
H1 (HA / H1N1)
H3 (HA / H3N2)
N1 (NA / H1N1)
N2 (NA / H3N2)
PA (H1N1)
PA (H3N2)
Number of tips
Year
0
1000
2000
3000
4000
2010.0
2012.5
2015.0
2017.5

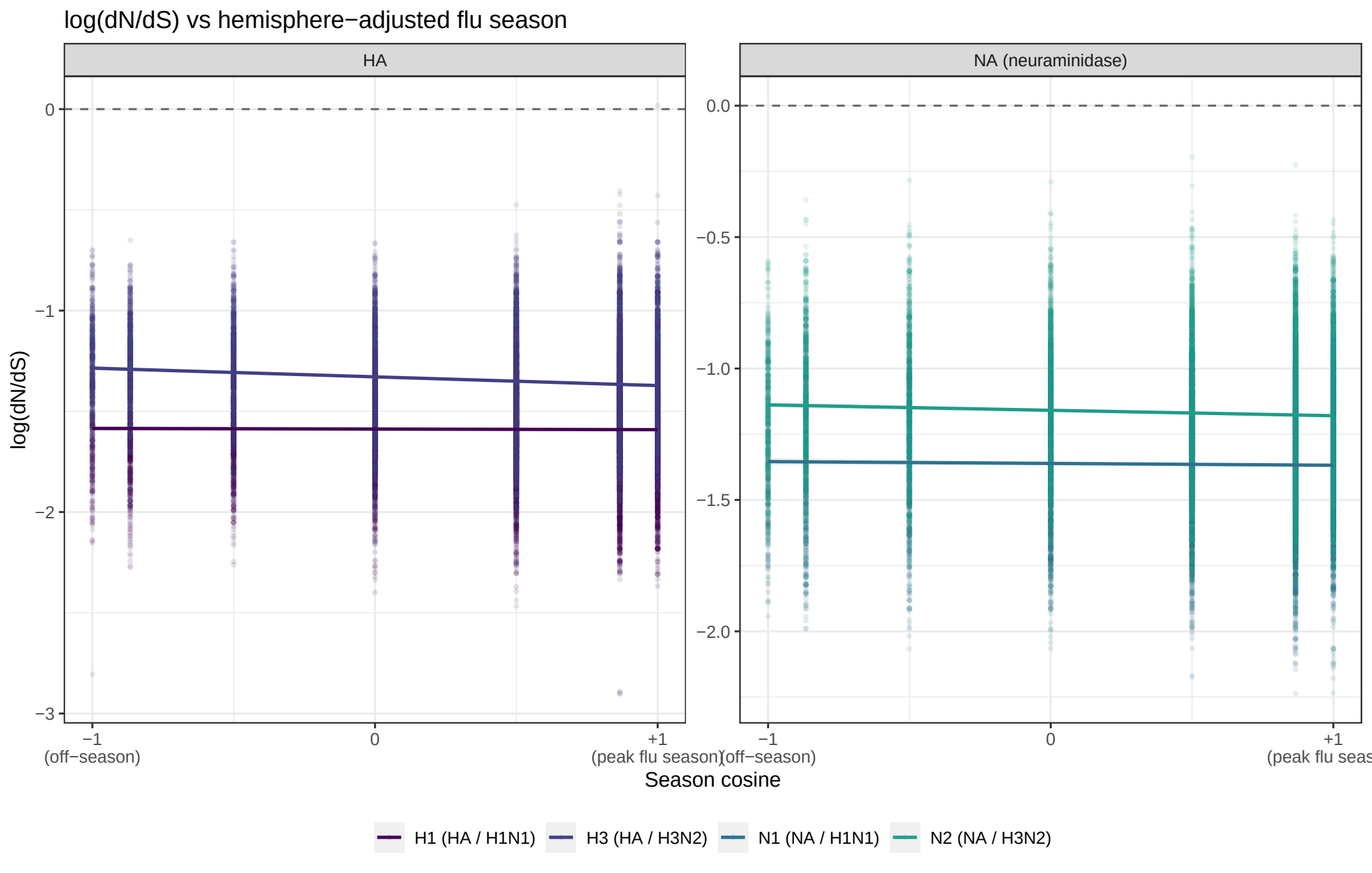

Distribution of log(dN/dS) by gene
Dashed line = dN/dS of 1 (neutral evolution)
Density
1.5
1.0
0.5
0.0
−3
−2
−1
0
log(dN/dS)
H1 (HA / H1N1)
H3 (HA / H3N2)
N1 (NA / H1N1)
N2 (NA / H3N2)


log(dN/dS) over time
H1N1
H3N2
0
−1
−2
−3
log(dN/dS)
2010.0
2012.5
2015.0
2017.5
Year
H1 (HA / H1N1)
H3 (HA / H3N2)
N1 (NA / H1N1)
N2 (NA / H3N2)


log(dN/dS) vs ERA5 temperature
HA
NA (neuraminidase)
log(dN/dS)
0
−1
−2
−3
0.0
−0.5
−1.0
−1.5
−2.0
−40
−20
0
20
Monthly mean temperature (°C)
H1 (HA / H1N1)
H3 (HA / H3N2)
N1 (NA / H1N1)
N2 (NA / H3N2)


log(dN/dS) vs relative humidity
HA
NA (neuraminidase)
log(dN/dS)
0
−1
−2
−3
0.0
−0.5
−1.0
−1.5
−2.0
25
50
75
Relative humidity (%)
H1 (HA / H1N1)
H3 (HA / H3N2)
N1 (NA / H1N1)
N2 (NA / H3N2)


log(dN/dS) by month of sample collection
H1 (HA / H1N1)
H3 (HA / H3N2)
N1 (NA / H1N1)
N2 (NA / H3N2)
log(dN/dS)
Month
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec


log(dN/dS) vs hemisphere−adjusted flu season
HA
NA (neuraminidase)
log(dN/dS)
0
−1
−2
−3
0.0
−0.5
−1.0
−1.5
−2.0
−1
(off−season)
0
+1
(peak flu season)
Season cosine
H1 (HA / H1N1)
H3 (HA / H3N2)
N1 (NA / H1N1)
N2 (NA / H3N2)

Geographic distribution of log(dN/dS)